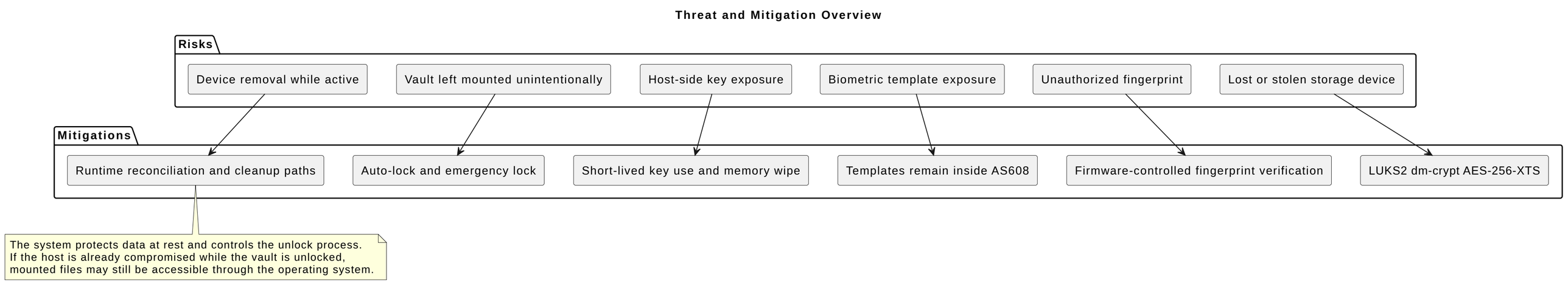
Threat and Mitigation Overview
Risks
Device removal while active
Vault left mounted unintentionally
Host-side key exposure
Biometric template exposure
Unauthorized fingerprint
Lost or stolen storage device
Mitigations
Runtime reconciliation and cleanup paths
Auto-lock and emergency lock
Short-lived key use and memory wipe
Templates remain inside AS608
Firmware-controlled fingerprint verification
LUKS2 dm-crypt AES-256-XTS
The system protects data at rest and controls the unlock process.
If the host is already compromised while the vault is unlocked,
mounted files may still be accessible through the operating system.